

WORKSHEET - NAPLAN Year 9

Number and Algebra Fractions, Decimals and Percentages (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. ☒ Number and Algebra, NAP-19142

The results from 400 metres running race final was recorded in the table below.

1st place	52.26 seconds
2nd place	52.62 seconds
3rd place	?
4th place	53.19 seconds

The time of the runner who came in 3rd place could be

- ☐ 52.57 seconds.
☐ 52.71 seconds.
☐ 53.21 seconds.
☐ 53.94 seconds.

2. ☒ Number and Algebra, NAP-19145

1 Australian dollar buys 0.60 US dollars

How many US dollars could be bought with 50 Australian dollars using this exchange rate?

3. ☒ Number and Algebra, NAP-48610

Which fraction has the same value as $2\frac{3}{5}$?

- $\frac{6}{5}$ $\frac{9}{5}$ $\frac{13}{5}$ $\frac{17}{5}$
☐ ☐ ☐ ☐

4. ☒ Number and Algebra, NAP-89737

Brandon has a money jar that contains 50 cent, 20 cent, 10 cent and 5 cent pieces only.

If he buys a pie worth \$2.65, what is the minimum amount of coins he can pay with?

- 5 7 8 9 11
☐ ☐ ☐ ☐ ☐

5. ☒ Number and Algebra, NAP-83790

Pete's seafood restaurant took in \$7703.40 on New Year's Eve.

Sam's Italian restaurant took in \$5091.75 on the same night.

How much more money did Pete's restaurant take in than Sam's on New Year's Eve?

\$

6. ☒ Number and Algebra, NAP-13287

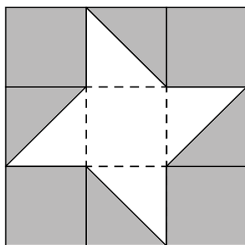
Scarlett has 8 cups of sugar. A muffin recipe uses $\frac{2}{3}$ of a cup of sugar for each muffin.

What is the greatest number of muffins Scarlett can make?

- 5 6 8 12 14
☐ ☐ ☐ ☐ ☐

7. ✕ Number and Algebra, NAP-07346

This shape is made up of 9 squares.



What fraction of the shape is white?

- ☐ $\frac{1}{5}$
☐ $\frac{1}{4}$
☐ $\frac{1}{3}$
☐ $\frac{1}{2}$

8. ✕ Number and Algebra, NAP-01626

Terry picked some pears from his orchard.

Exactly 45% of the pears picked were not good to eat.

What is the smallest total number of pears that he could have collected?

 pears

Worked Solutions

1. Number and Algebra, NAP-19142

**52.71 is slower than 2nd place (52.62)
but faster than 4th place (53.19).**

2. Number and Algebra, NAP-19145

$$\begin{aligned}\text{Number of USD} &= 50 \times 0.60 \\ &= \$30 \text{ USD}\end{aligned}$$

3. Number and Algebra, NAP-48610

$$\begin{aligned}2\frac{3}{5} &= \frac{10}{5} + \frac{3}{5} \\ &= \frac{13}{5}\end{aligned}$$

4. Number and Algebra, NAP-89737

Minimum number of coins would be:

$$5 \times 50 \text{ c}$$

$$1 \times 10 \text{ c}$$

$$1 \times 5 \text{ c}$$

∴ A minimum of 7 coins is needed.

5. Number and Algebra, NAP-83790

$$\begin{aligned}\$7703.40 - \$5091.75 \\ = \$2611.65\end{aligned}$$

6. Number and Algebra, NAP-13287

Greatest number of muffins

$$= 8 \div \frac{2}{3}$$

$$= 8 \times \frac{3}{2}$$

$$= 12$$

7. Number and Algebra, NAP-07346

Dividing the shape into 18 triangles,

$$\text{White triangles} = \frac{6}{18} = \frac{1}{3}$$

8. Number and Algebra, NAP-01626

$$45\% = \frac{9}{20}$$

The minimum number of pears that could be collected is the smallest number that can be

be multiplied by $\frac{9}{20}$ and equal a whole

number.

$$\therefore \text{Smallest number of pears} = 20$$